

Project Context Specification
Stock Management System

PART 1: PROJECT OVERVIEW

Project Name*: Stock Management System
Application Display Name*: Inventory Manager

Project Description*:
The Stock Management System centralizes inventory operations for an organization. It manages products, suppliers, purchase orders, stock receipts, stock issues, inventory adjustments, warehouses, and reporting. The system replaces manual stock registers, provides real-time inventory visibility, reduces stock losses, and supports better purchasing decisions.

PART 2: SYSTEM USERS

User	Type	Responsibilities
System Administrator	Internal	Manage users, roles, settings, audit logs.
Storekeeper	Internal	Manage products and inventory transactions.
Procurement Officer	Internal	Manage suppliers and purchase orders.
Department Manager	Internal	Approve requests and view reports.
Auditor	Internal	Read-only access to reports and audit logs.

Public access: No

PART 3: FUNCTIONAL AREAS

- User Management
- Product Management
- Supplier Management
- Inventory Management
- Stock Transactions
- Purchase Management
- Reports & Dashboard

PART 4: FEATURE DETAILS

User Management

Use Case: Create User

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

1. User opens module.
2. System displays interface.
3. User enters required information.
4. System validates data.
5. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Use Case: Manage Roles

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

6. User opens module.
7. System displays interface.
8. User enters required information.
9. System validates data.
10. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Product Management

Use Case: Add Product

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

11. User opens module.
12. System displays interface.
13. User enters required information.
14. System validates data.
15. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Use Case: Update Product

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

16. User opens module.
17. System displays interface.
18. User enters required information.
19. System validates data.
20. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Use Case: Deactivate Product

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

21. User opens module.
22. System displays interface.
23. User enters required information.
24. System validates data.
25. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Supplier Management

Use Case: Register Supplier

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

26. User opens module.
27. System displays interface.
28. User enters required information.
29. System validates data.
30. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Use Case: Update Supplier

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

31. User opens module.
32. System displays interface.
33. User enters required information.
34. System validates data.
35. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Inventory Management

Use Case: View Stock

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

36. User opens module.
37. System displays interface.
38. User enters required information.
39. System validates data.
40. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Use Case: Perform Stock Count

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

41. User opens module.
42. System displays interface.
43. User enters required information.
44. System validates data.
45. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Stock Transactions

Use Case: Record Stock In

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

46. User opens module.
47. System displays interface.
48. User enters required information.
49. System validates data.
50. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Use Case: Record Stock Out

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

51. User opens module.
52. System displays interface.
53. User enters required information.
54. System validates data.
55. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Use Case: Transfer Stock

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

56. User opens module.
57. System displays interface.
58. User enters required information.
59. System validates data.
60. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Use Case: Adjust Stock

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

61. User opens module.
62. System displays interface.
63. User enters required information.
64. System validates data.
65. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Purchase Management

Use Case: Create Purchase Order

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

66. User opens module.
67. System displays interface.
68. User enters required information.
69. System validates data.
70. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Use Case: Receive Goods

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

71. User opens module.
72. System displays interface.
73. User enters required information.
74. System validates data.
75. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Reports & Dashboard

Use Case: Generate Reports

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

76. User opens module.
77. System displays interface.
78. User enters required information.
79. System validates data.
80. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

Use Case: View Dashboard

Preconditions: Authenticated user with required permission.

Postconditions: Relevant records are saved and audit logs updated.

Main Flow:

81. User opens module.
82. System displays interface.
83. User enters required information.
84. System validates data.
85. System saves changes and confirms success.

Alternative Scenario: Validation fails, duplicate data exists, insufficient permissions, or business rules are violated.

PART 5: SPECIAL REQUIREMENTS

File Uploads: Yes

Areas: Products, Suppliers, Purchase Orders.

Formats: PDF, PNG, JPG, JPEG.

Maximum size: 10 MB.

Data Export: PDF and Excel for inventory, purchases, suppliers, transactions, and dashboard reports.

Search & Filtering:

Products by SKU, name, category.

Suppliers by name.

Transactions by date, warehouse, type.

Purchase Orders by supplier and status.

PART 6: PREFERENCES

Language: English and French

Colour Scheme: Afriland Corporate Colours

Additional Notes:

- Responsive web application
- Mobile friendly
- Email notifications for low stock
- Role-based access control
- Complete audit trail
- Automatic session timeout

PART 7: ADVANCED INFORMATION

Data Relationships

- One Category -> Many Products
- One Supplier -> Many Products
- One Supplier -> Many Purchase Orders
- One Purchase Order -> Many Purchase Items
- One Product -> Many Stock Transactions
- One Warehouse -> Many Products
- One User -> Many Transactions

Business Rules

- SKU must be unique.
- Stock cannot become negative.
- Products with transaction history cannot be deleted.
- Every stock adjustment requires a reason.
- Only Procurement Officers approve purchase orders.
- Only Administrators manage users.
- Automatic low-stock alerts when quantity falls below minimum.
- Every transaction is logged for auditing.